1. Representation leads for binary point decomposition

Campaign `research/representation_leads_20261109/`, 2026-11-09. Exact derivations in native Rust (`derive/`), no timings, no experiment runs. **Every number is an unverified assertion:** no pcc claim was minted, and the `research-kb` server was unreachable, so novelty rests on the lead atlas `--match` and one literature search.

1.1. Why these representations

An atlas finding on ECC2K-130 (KN-FIND-aa2efc, unreviewed) gives the product law relations × targets × attempts = m·2^{131}. With free attempts the floor is $2^{68.58}$ at m = 3 (above rho's $2^{60.81}$), $2^{56.40}$ at m = 4, $2^{48.44}$ at m = 5 and $2^{42.85}$ at m = 6. A representation therefore matters only if it makes one attempt fit roughly 2^{12} (m = 5) to 2^{18} (m = 6) rho-step equivalents: the leads aim at m ≥ 5.

1.2. R1′: quadratised chains (line norm form)

For a merge $P_a + P_b = U$ on $y^2 + xy = x^3 + ax^2 + b$, the line through P_a, P_b and $-U$ gives $e_1 = \lambda^2 + \lambda + a$, $e_2 = \mu$, $e_3 = \mu^2 + b$ (unit-tested on 1000 random pairs). Keeping $\mu = e_2$ replaces the cubic $S_3(x_a, x_b, x_U)$ by two quadratics: $x_a x_b + x_U(x_a + x_b) = \mu$ and $\mu x_U + x_U^2(x_a + x_b) + \mu^2 + b = 0$. The descended chain then has degree 2 at every m.

Size (exact). Against the confound control (w = $x_a x_b$ on the same merges), $\mu = e_2$ has the same variable count and the same set of distinct monomials, and 30–40 % more terms (figure below). On written size it is a relabelled generic degree reduction.

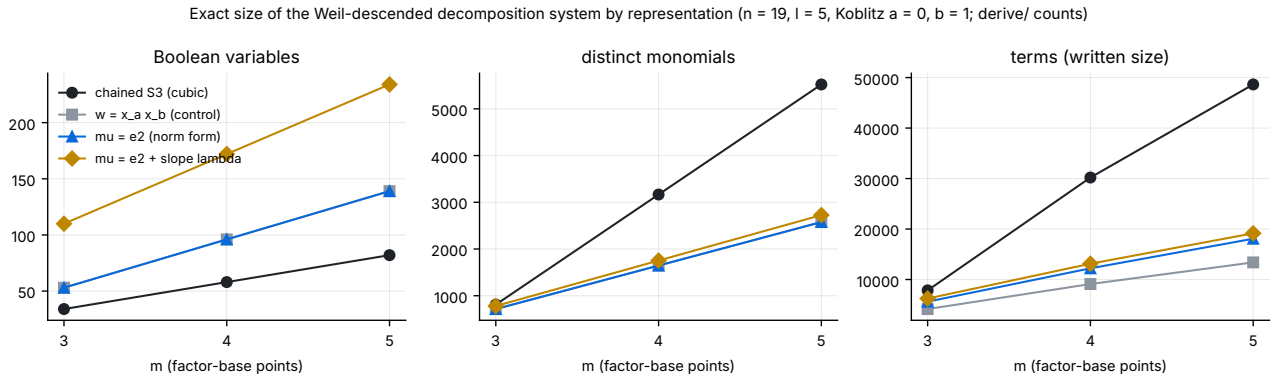


Figure 1: Exact sizes at n = 19, l = 5 (Koblitz a = 0, b = 1). All 288 built systems satisfy their planted relation; single-bit flips are detected in ≥ 31/32 cases per system.

The slope λ. Over GF(2^n), $\lambda^2 + \lambda = c$ is solvable exactly when Tr(c) = 0. An exhaustive census of every root triple of $S_3(x_1, x_2, X) = 0$ (n = 11, 13; four curves each) finds only two classes: all three abscissae on E with λ in the field, or all three on the twist with none. Mixed triples never occur, and every sampled on-E triple lifts to a true sum. In a chain the merges share abscissae and the last contains x_R on E, so the trace equations are **implied**; the atlas already records the trace linear equation (KR-IC-f5c584). λ is eliminable (one linear equation per merge).

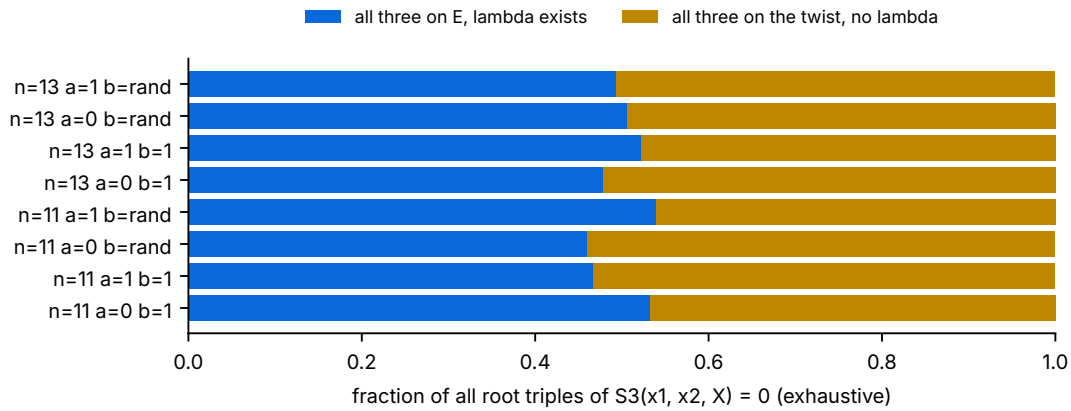


Figure 2: Exhaustive census of S_3 root triples (results/census.jsonl).

Prior art for λ. The trace map behind this is the Lenstra–Kosters homomorphism $E(k) \to \mathbb{F}_2$, $P \mapsto \mathrm{Tr}\left(\frac{x(P)+a_2}{a_1^2}\right)$ with kernel $2E(k)$ (2013), derived from the same line identity $x_1 + x_2 + x_3 = \lambda^2 + a_1\lambda + a_2$; it is the known explanation of the first fall degree 2 of S_3 systems (Kosters, ECC 2015 slides; atlas KR-IC-f5c584). The census is an exhaustive confirmation at n = 11, 13, not a new fact.

Open question (gated). Whether $\mu = e_2$, or the explicit implied trace equations, lowers the closure refutation degree relative to its matched control. `PROTOCOL.md` (frozen 2026-11-14) fixes four arms in two matched pairs with identical variable sets, eight cells, 16 exhaustively unsatisfiable and 4 planted draws per cell, and the instrument `examples/replead_closure.rs` (the routines of `examples/dreg_closure.rs`). The 640 systems are pinned by sha256; running needs the checkpoint token.

1.3. R3: odd power-sum coordinates

In characteristic 2, $p_k = \sum_i x_i^k$ is slot-separable on a subspace. Exact linear algebra over GF(2^{13}) and GF(2^{17}), three (b, x_R) draws each: $p_1 S_3$ lies in $F[p_1, p_3, p_5, \ldots]$, which is the generic pair-table form; $p_1^j S_4$ does not, for any j ≤ 8, on all six draws. S_4 is in the elementary-symmetric span, and degree-12 positive controls in three variables are recovered (j = 0), so the method sees such elements. Closed in scope.

1.4. R2: membership as a Frobenius chain modulo N

$N = \prod(X - x_i)$ divides the subspace polynomial L_V iff all roots lie in V; tracking $R_k = X^{2^k} \bmod N$ with $R_k = R_{k-1}^2 \bmod N$ is bilinear, so membership is degree 2 for all m at l·m·n auxiliary bits.

Closed: dominated inside the Huang–Kosters–Yang–Yeo framework (arXiv:1505.02532, Theorem 1.1, read 2026-11-14). For a zero-dimensional radical big-field system $\mathcal{F}$ in M variables with at most s solutions over the closure, the last fall degree of its full Weil descent is at most $\max(\tau(\max(d_{\mathcal{F}}, \deg \mathcal{F}, (M+1)s, 1), q, M), M \cdot \tau(2s, q, 1), q)$ with $\tau(r, c, t) = \max\Big(\Big\lfloor 2t(c-1)\Big(\log_{c(\frac{r}{2t})} + 1\Big)\Big\rfloor, 0\Big)$; their Section 4.4 says, without proof, that the same should hold with added constraints of Frobenius weight at most $\tau(2s, q, 1)$, "such as linear subspace constraints". L_V is linearized (weight 1), so the standard system (S_{m+1} plus $L_V(x_i) = 0$, M = m) already sits in that scope; the squaring chain lowers only the ordinary degree, which the bound does not use, and adds l·m big-field variables, which enter it linearly. Scope: the subspace case is an unproved remark there, and $d_{\mathcal{F}}$ is bounded for neither system.

1.5. R4 and R5

R4 (uncorrelated slots for the n = 83, m = 10 gate) is cell G6 of PR #2679 and waits for the checkpoint token. R5 (mixed chains) is the R1′ construction itself: auxiliaries only on merges whose S_3 is cubic.

1.6. Scope and limits

- Sizes are at toy n (11–19); no solving degree, timing or cost is measured.
- Census and power-sum results are exact on the stated fields and draws only.
- Nothing here changes a leaderboard, benchmark or validation row.